HIV INFECTION

HIV is carried in blood, semen, vaginal secretions, and breast milk. It is passed on when infected fluids enter the body. HIV infects cells with structures called CD4 molecules on their surface. These CD4+ cells include lymphocytes, which fight infection. The virus multiplies rapidly in CD4+ cells, destroying them in the process. If HIV goes untreated, the number of CD4+ lymphocytes eventually falls so low that the immune system is severely weakened.

AIDS

HIV can be identified by specific blood or fluid tests. Being HIV positive may lead to AIDS-related illnesses, especially opportunistic infections, caused by organisms that are harmless to healthy people but dangerous to those with reduced immunity; one example is infection by *Candida albicans*, which causes thrush. People who have AIDS may also develop various types of cancer, notably Kaposi's sarcoma.

1 FREE HIV PARTICLE The capsid contains two strands of RNA, each carrying a set of genes for the virus. The spikes on the surface are proteins called gp120 antigens that enable the virus to "dock" on the surface of CD4+ cells.

2 BINDING AND INJECTION The gp120 binds to CD4 molecules and then to co-receptors on the cell surface. The virus fuses with the cell, penetrating the surface. The capsid releases the viral RNA.

3 REVERSE TRANSCRIPTION The virus releases reverse transcriptase into the cell. The enzyme copies the single strands of viral RNA as double-stranded DNA.

4 INSERTION OF VIRAL DNA Viral DNA enters the nucleus and is incorporated into the cell's DNA. The cell produces mRNA, which has instructions for making new proteins, including HIV proteins.

5 CREATION OF PROTEINS In the cytoplasm the mRNA is "read" and chains of HIV proteins and viral RNA are made. These molecules become the components of new HIV particles.

6 NEW HIV CREATED HIV constituents gather at the cell wall. An immature virus forms and breaks away from the cell. Enzymes within the virus bring about changes resulting in a mature virus.